DPRE in dim $1+d$

· Easy picture: random walk$^{(1)}$ in random potential$^{(2)}$.

· (1) on $(\Omega_{traj}, \mathcal{F}, P_x)$, the random sequence

· $S = (S_n)_{n \geq 0} : \Omega_{traj} \to (\mathbb{Z}^d)^{\mathbb{N}_{\geq 0}}$ equipped with cylinders,

· $P_x(S_0 = x) = 1$, $P_x(S_n - S_{n-1} = \pm e_j) = (2d)^{-1}$, $j = 1, \dots, d$.

· E_x denotes P_x-expectation,

$P_0 =: P$.

· (2) on $(\Omega, \mathcal{G}, \mathbb{P})$, the random sequence

· $\omega = (\omega(n,x) : n \in \mathbb{N}, x \in \mathbb{Z}^d) : \Omega \to \mathbb{R}^{\mathbb{N} \times \mathbb{Z}^d}$,

where (i) $\omega(n,x)$ are iid

(ii) $\mathbb{E}\left[e^{\beta \omega(n,x)}\right] < \infty \;\; \forall \beta \in \mathbb{R}$,

$\Rightarrow$ in particular, all moments of ω exist.

· the <u>polymer measure</u>: measure on path space $(\mathbb{Z}^d)^{\mathbb{N}_{\geq 0}}$

$$P_n^{\beta,\omega}(d\underset{\sim}{x}) := \frac{1}{Z_n(\omega,\beta)} e^{\beta H_n^{\omega}(\underset{\sim}{x})} P(d\underset{\sim}{x}),$$

"directed" in time,

- $\beta > 0$ (wlog, else take $-\omega$) inverse temperature,

- $H_n^\omega(\underset{\sim}{x}) := \sum_{j=1}^{n} \omega(j, \underset{\sim}{x}_j)$ Hamiltonian,

- $$Z_n = Z_n(\omega, \beta) := E\left[e^{\beta \sum_{j=1}^{n} \omega(j, S_j)}\right]$$

$$= \sum_{\underset{\sim}{x} \text{ SRW paths}} (2d)^{-n} e^{\beta H_n^\omega(\underset{\sim}{x})},$$

the normalizing factor in the Radon-Nikodym derivative, which we call the partition function.

Note well that Z_n is random still in ω: physicists call this a quenched object.

To wit, the language we use suggests physical motivation for these definitions. If indeed we view ω as an energy field, or pockets of disorder,

- H_n measures the energy along a given path $\underset{\sim}{x}$,

- Z_n measures the total spread of energy across the system.*

Then $P_n^{\beta,\omega}$ reweights diffusive paths, emphasizing trajectories that "pick up" more energy.

ie "the polymer is attracted to sites where energy is higher, repulsed by sites where it is lower."

* Originally, Z_n came about in physical chemistry as a way to count particles in a system distributed over energy levels according to the <u>Boltzmann distribution</u>:

- if n_0 is the # of particles in the ground state, n_i " " in i^{th} excited state,

then $$\frac{n_i}{n_0} = \frac{1}{e^{\varepsilon_i/kT}} = \frac{1}{e^{\beta\varepsilon_i/k}}.$$

Boltzmann constant

Then if N is the total volume of the system, we can write
$$\frac{N}{n_0} = \sum_{i=0}^{\infty} e^{-\beta\varepsilon_i/k} =: Z_N,$$

so that Z_N is a measure of the degree to which particles are "partitioned" among energy levels, hence its name.

- The physical intuition for why $P_n^{\beta,w}$ is a Gibbs measure is that we expect the future trajectories of an energy-attractive polymer, conditional on where it is now, to be independent of past sites visited; and the result that (Hammersley-Clifford) "every probability measure satisfying a Markov property is a Gibbs measure".

- <u>DPRE and last passage percolation</u> Fix n.

  - You may already have noticed that $\beta \downarrow 0$ recovers the random walk measure.
  - What about $\beta \uparrow \infty$?

$$\begin{cases} \dfrac{\log Z_n(\omega, \beta)}{\beta} \longrightarrow \max_{\underset{\sim}{x}} H_n(\underset{\sim}{x}), \\ P_n^{\beta,\omega}(\operatorname{argmax}_{\underset{\sim}{x}} H_n(\underset{\sim}{x})) \longrightarrow 1, \end{cases}$$

that is, we recover from Z_n the passage time of a last passage percolation problem (viewing the ω as costs), and $P_n^{\beta,\omega}$ concentrates on the geodesics thereof.

Hence the parameter β lets us interpolate between

- a diffusive, "high temperature" regime, where ω perturbs the random walk only slightly, and
- a localizing, "low temperature" behavior, where paths cluster in favorable energic corridors, and the global shape of the polymer paths are driven by ω to look quite different.

Studying this "phase transition", from a weak- to strong-disordered regime, has been of interest to many problems in the theory of disordered systems, for example,

<u>stochastic heat flow</u>.
(random)

· <u>DPRE and heat flow: heuristics</u> Fix $d = 1$.

· Let $\beta = 0$, that is, consider the random walk $(S_n)_{n \geq 0}$.

· We know by CLT that under diffusive scaling

ξ_i iid, $\sum_{i=1}^{n} \xi_i$,

$P(\xi_i = \pm 1) = \frac{1}{2}$

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} \xi_i \xrightarrow[n \uparrow \infty]{\mathcal{D}} W_t \sim \mathcal{N}(0, t)$$

the random walk converges in distribution to Brownian motion.

· We also know that, for $f \in C_b^2(\mathbb{R})$, the function

$$v(t, x) := E\left[f(x + W_t)\right]$$

solves the heat equation

$$\begin{cases} \partial_t v = \frac{1}{2} \Delta v, \\ u(0, x) = f(x); \end{cases}$$

(This was a homework problem in Assignment #3)

ie "classical heat flow is the average over paths of diffusive particles".

Now, what if these particles are "kicked" by small forces in their environment that perturb their diffusive behavior? That is, is there a meaningful way to solve the <u>stochastic heat equation</u>

$$\begin{cases} \partial_t u = \frac{1}{2}\Delta u + f(u)\dot{W}, \\ u(0,x) = g(x) \end{cases}$$

(where $f \in C^0(\mathbb{R})$ and $\dot{W}(t,x)$ is (time-space) <u>white noise</u>, which has rigorous definition as either a random $\underbrace{\text{distribution}}$

= linear functional on test functions, ie generalized function,

or a Gaussian family indexed by $\mathcal{B}(\mathbb{R}_{\geq 0} \times \mathbb{R}^d)$, but can here be viewed as just the formal derivative of Brownian motion $W(t) = \int_0^t \dot{W}(s)$) ?

We are pessimistic, since the SHE is a nonlinear SPDE driven by a very singular noise (Brownian motion is notorious for being nowhere differentiable along $\mathbb{P}$-a.e. path $t \mapsto W_t(\omega)$).

† In fact, even if we take $f = 1$ to linearize the SHE (call this the additive SHE), a heuristic computation using the Duhamel principle for $\underbrace{\text{inhomogeneous}}$ PDEs shows that, for $d \geq 2$, u is at best distribution-valued:

$\partial_t u = \frac{1}{2}\Delta u + \xi(t,x)$

$\uparrow$ external heat source

~ "family of homogeneous PDEs with updated initial condition at every time slice"

† the <u>mild solution</u> u to the additive SHE

$$\begin{cases} \partial_t u = \frac{1}{2}\Delta u + \dot{W} \\ u(0,x) = g(x) \end{cases}$$

has Duhamel form

convolution with initial condition + with external heat source

$$u(t,x) = \int_{-\infty}^{\infty} G(t, x-y)\, g(y)\, dy + \int_{-\infty}^{\infty}\int_0^t G(t-s, x-y)\, dW(s,y),$$

where $G(t,x) = \frac{1}{(2\pi t)^{d/2}} e^{-\|x\|^2/2t}$ is the d-dim heat kernel.

For the Itô integrals above to make sense and define a random function, we need

$$E\left[u(t,x)^2\right] = \int_{-\infty}^{\infty}\int_0^t G(t-s, x-y)^2\, ds\, dy < \infty, \quad \forall t.$$

For $d = 1$, we actually get

$$\int_0^t\int_{-\infty}^{\infty} \frac{1}{2\pi(t-s)} e^{\frac{(x-y)^2}{t-s}}\, dy\, ds = \int_0^t \frac{1}{2\sqrt{\pi(t-s)}}\, ds = \sqrt{\pi t} < \infty,$$

reflecting the fact that the solution theory in $1+1$ dim is more robust. But for $d = 2$,

$$\int_0^t\int_{-\infty}^{\infty} \frac{1}{(2\pi(t-s))^2} e^{\frac{2\|x-y\|^2}{t-s}}\, dy\, dt = \int_0^t \frac{1}{8\pi(t-s)}\, ds = \infty.$$

So u must be interpreted as a random distribution instead, and it becomes hard even to say what it means to solve the SHE.

Caravenna, Sun, and Zygouras (2023): in $d = 2$, and for β in some "critical window" ** (discuss later),

"the diffusively rescaled, averaged point-to-point partition function converges in finite-dimensional distributions to the same limit as do solutions to the mollified SHE."

$\downarrow$ smooth out spatially

(convolve $\dot{W}$ with smooth kernel)

ie define

$$Z^{\beta,\omega}_{M,N}(w,z) := E\left[e^{\beta\sum_{n=M+1}^{N-1}\overset{***}{(\omega(n,S_n) - \lambda(\beta))}} \mathbb{1}\{S_N = z\} \,\Big|\, S_M = w\right]$$

"point to point partition function",

then

Lebesgue meas. on $\mathbb{R}^2 \times \mathbb{R}^2$

$$Z^\beta_N = \left(Z^\beta_{N;s,t}(dx,dy) := \frac{N}{4} Z^{\beta,\omega}_{[\![Ns]\!],[\![Nt]\!]}\left([\![\sqrt{N}x]\!],[\![\sqrt{N}y]\!]\right) dx\,dy\right)_{0\le s\le t<\infty}.$$

periodicity

$$\underbrace{\text{for } s\in\mathbb{R},\ [\![s]\!] := 2\lfloor s/2\rfloor \in 2\mathbb{Z} \qquad \text{for } x\in\mathbb{R}^2,\ [\![x]\!] := \text{closest point in } \mathbb{Z}^2_{\text{even}} := \{(z_1,z_2)\in\mathbb{Z}^2 : z_1+z_2 \text{ even}\}}_{\text{even approximations, for parity issues.}}$$

"diffusively rescaled partition functions"

= family of random measures on $\mathbb{R}^2\times\mathbb{R}^2$.

Under the topology of vague convergence,

$$\left(\mu_N \to \mu \iff \int\phi\mu_N \to \int\phi\mu \quad \forall\phi\in C_c(\mathbb{R}^2\times\mathbb{R}^2)\right),$$

$$Z^\beta_{N;s,t}(\varphi,\psi) \xrightarrow{d} \mathscr{Z}_{s,t}(\varphi,\psi), \quad 0\le s\le t\le\infty$$

unique, and shared by limit of u_δ as $\delta\downarrow 0$

solutions to mollified SHE.

$$:: \iint_{\mathbb{R}^2\times\mathbb{R}^2} \varphi(x)\psi(y) Z^\beta_N(dx,dy)$$ "averaging over paths".

Many technical details are left out for time's sake, but the spirit should feel familiar:

- taking the diffusive limit of a random walk, then averaging over paths, we get a solution to the classical heat equation;

- taking the diffusive limit of the point to point partition function averaged over paths, we "get" a solution to the stochastic heat equation.

This solution is called the 2d critical stochastic heat flow, reflecting our choice of $\beta > 0$ in an appropriate window "between weak and strong noisiness".

To explain what I mean, we need to examine the partition function more closely.

First we define a <u>shift operator</u> $\vartheta_{i,x}$ on the space of environments

$$\omega(t,y) \mapsto (\vartheta_{i,x}\,\omega)(t,y) = \omega(i+t, x+y)$$

"field of environment variables seen from (i,x)".

Then for $n, m \geq 1$, $x \in \mathbb{Z}^d$, the random variable

$$\begin{aligned} Z_m \circ \vartheta_{n,x}(\omega) &= Z_m(\vartheta_{n,x}\,\omega\,;\beta) \\ &= E_x\left[e^{\beta\sum_{t=1}^{m}\omega(t+n, S_t)}\right] \end{aligned}$$

is the partition function of length m started at (n,x).
By the Markov property of $(S_n)_{n\geq 0}$,

$$\text{if } \mathcal{F}_n := \sigma(S_t : t \leq n),$$

we can write

$$\begin{aligned} Z_m \circ \vartheta_{n,x}(\omega) &= E\left[e^{\beta(H_{n+m}(S) - H_n(S)}\,\middle|\,\mathcal{F}_n\right] \quad \text{on } \{S_n = x\} \\ &= E\left[e^{\beta(H_{n+m}(S) - H_n(S))}\,\middle|\,S_n = x\right]. \end{aligned}$$

Then

$$\begin{aligned} Z_{n+m} &= E\left[e^{\beta H_{n+m}(S)}\right] \\ &= E\left[e^{\beta H_n(S)}\,E\left(e^{\beta(H_{m+n}(S) - H_n(S))}\,\middle|\,\mathcal{F}_n\right)\right] \\ &= E\left[e^{\beta H_n(S)}\,Z_m \circ \vartheta_{n,S_n}(\omega)\right] \\ &= Z_n\,E^{\beta}\left[Z_m \circ \vartheta_{n,S_n}\right], \quad \text{since } \omega \text{ and } \vartheta\omega \text{ have same law.} \end{aligned}$$

(wrt polymer measure)

Call this the <u>Markov property</u> of Z_n. This gives a first intuition for why Z_n is a good candidate solution.

Now, how do we study the dependence of Z_n on β? Consider

$$\cdot \; p_n = p_n(\omega, \beta) := \frac{1}{n} \log Z_n(\omega, \beta) \quad \underline{\text{quenched free energy}},$$

specific, finite volume

$$\cdot \; \lambda(\beta) := \log \mathbb{E}\left[e^{\beta \omega(n,x)}\right] \quad \underline{\text{annealed free energy}}.$$

· <u>Thm</u> (Comets). $\lim_{n\to\infty} p_n(\omega, \beta)$ exists $\mathbb{P}$-a.e., and in fact

$$p_n(\omega, \beta) \xrightarrow[n]{a.e.} p(\beta) := \sup_n \frac{1}{n} \mathbb{E}\left[\log Z_n(\omega, \beta)\right]$$

In particular, $p(\beta)$ does not depend on ω! Call it the <u>infinite volume</u>, <u>specific free energy</u>

One way to measure disorder strength is via the <u>annealed bound</u>:

Jensen's

$$\frac{1}{n} \mathbb{E}\left[\log Z_n\right] \leq \log \mathbb{E}(Z_n)^{\frac{1}{n}} = \lambda(\beta) \quad \forall n$$

$$\Rightarrow \quad p(\beta) \leq \lambda(\beta).$$

We ask: is this bound sharp?

Intuitively, Jensen's is closer to equality when the convex function fluctuates little,
and also correlations in summands increase fluctuations of the sum.

Here, our sum is an average over walk paths, so we expect that

- the annealed bound is sharp when $H_n(S)$ is uncorrelated over walk paths S (weak ω),
- the annealed bound is crude when $H_n(S)$ is correlated " " (strong ω).

· Thm (critical temperature 1). $\exists \beta_c \in [0, \infty]$ such that

$$\begin{cases} p(\beta) = \lambda(\beta), & 0 \leq \beta \leq \beta_c \\ p(\beta) < \lambda(\beta), & \beta > \beta_c. \end{cases}$$

· Call · $\{\beta : p = \lambda\}$ the high T, weak - disordered
· $\{\beta : p < \lambda\}$ the low T, strong - disordered
regions.

It turns out to be grueling to find β_c, particularly for larger d. A second approach turns out to be nicer, in so far as it is amenable to martingale methods!

*** explains the $\lambda(\beta)$ in CSZ

· First, let's define

$$W_n := Z_n(\omega;\beta)\, e^{-n\lambda(\beta)}$$

"normalized partition function"

since $\mathbb{E}(W_n) = 1$.

· If we fix $\underset{\sim}{x}$ a random walk path, then each $H_k(\underset{\sim}{x})$ is a sum off iid $\omega(i, \underset{\sim}{x}_i)$ on $(\Omega, \mathcal{G}, \mathbb{P})$,

and $e^{\lambda(\beta)} = \mathbb{E}\left[e^{\beta\omega}\right]$ is the mgf.

Then

$$\zeta_n(\underset{\sim}{x}) := e^{\beta H_n(\underset{\sim}{x}) - n\lambda(\beta)} = \frac{e^{\beta\sum_{i=1}^{n}\omega(i,\underset{\sim}{x}_i)}}{mgf_\omega(\beta)^n}$$

is martingale,

with respect to $\left(\mathcal{G}_n = \sigma(\omega(j,x) : j \leq n,\ x \in \mathbb{Z}^d)\right)_{n\in\mathbb{N}}$

$\Rightarrow \sum_{\underset{\sim}{x}} c_{\underset{\sim}{x}}\, \zeta_n(\underset{\sim}{x})$ is mg (finite linear combo)

$\Rightarrow E[\zeta_n] = W_n$ is mg.

This is a much stronger statement than $\mathbb{E}(W_n) = 1$, and makes Z_n much easier to study than $\log Z_n$.

See that $\zeta_n(\underset{\sim}{x}) \longrightarrow 0$ $\mathbb{P}$ - a.e., $\forall \underset{\sim}{x}$:

Pf. $\cdot$ $\frac{1}{n} \log \zeta_n(\underset{\sim}{x}) = \beta \frac{1}{n} \sum_{i=1}^{n} \omega(i, \underset{\sim}{x}_i) - \log \mathbb{E}[e^{\beta\omega}]$.

$$\text{SLLN} \Rightarrow \log \zeta_n(\underset{\sim}{x})^{\frac{1}{n}} \xrightarrow[n]{a.e.} \beta \mathbb{E}\omega - \log \mathbb{E}[e^{\beta\omega}]$$

$$\zeta_n(\underset{\sim}{x})^{\frac{1}{n}} \xrightarrow[n]{a.e.} \frac{e^{\beta \mathbb{E}\omega}}{\mathbb{E}[e^{\beta\omega}]} < 1 \text{ by Jensen's}$$

$$\Rightarrow \zeta_n(\underset{\sim}{x}) \xrightarrow[n]{a.e.} 0, \text{ since } \sqrt[n]{a} \xrightarrow[n]{} 1 \ \forall a > 0.$$

$\square$

But $\mathbb{E}[\zeta_n(\underset{\sim}{x})] = W_n$ might converge to something positive:

- $\zeta_n(\underset{\sim}{x})$ decays exponentially
  ($\mathbb{P}(\zeta_n(\underset{\sim}{x}) > t) \approx e^{-n I(t)}$ by large deviations theory,
  $I(x) = x \log x + (1-x)\log(1-x) + \log 2$),
- but the sum over $(2d)^n$ paths in $\mathbb{E}$ grows exponentially.

Upshot: instead of studying criticality based on sharpness of $p \leq \lambda$, we can look at positivity of W_∞, then see if these frameworks agree.

· Thm $W_\infty := \lim_{n\to\infty} W_n$ exists $\mathbb{P}$-a.e., and either

$$\begin{cases} \mathbb{P}(W_\infty > 0) = 1 & \text{or} \\ \mathbb{P}(W_\infty = 0) = 1, \end{cases}$$

ie we have dichotomy!

Pf. W_n is a nonnegative martingale, so W_∞ exists $\mathbb{P}$-a.e. by Doob.

To prove dichotomy, it is enough to show that

$$\{W_\infty = 0\} \in \mathcal{T} := \bigcap_{n\geq 1} \mathcal{T}_n, \quad \mathcal{T}_n = \sigma(\omega(j,x) : j \geq n)$$

tail σ-field.

$\forall n$ Markov property of Z_n

$$\Rightarrow W_{n+m} = E(\zeta_n W_m \circ \vartheta_{n,S_n}),$$

so
$$W_\infty = \lim_{m\to\infty} W_{n+m}$$
$$= E[\zeta_n \lim_{m\to\infty} W_m \circ \vartheta_{n,S_n}] \quad (E \text{ is finite } \Sigma)$$
$$= \underbrace{W_n}_{>0} \sum_{x\in\mathbb{Z}^d} P_n^{\beta,\omega}(S_n = x)\, W_\infty \circ \vartheta_{n,x}$$

$$\Rightarrow \{W_\infty = 0\} = \bigcap_{x\in\mathbb{Z}^d : P(S_n = x)>0} \{W_\infty \circ \vartheta_{n,x} = 0\} \in \mathcal{T}_n.$$

n was arbitrary $\Rightarrow \{W_\infty = 0\} \in \mathcal{T}$.

· **Prop** (critical temperature 2). $\exists \overline{\beta}_c \in [0, \infty]$ such that

$$\begin{cases} W_\infty > 0 \text{ a.e. if } \beta \in \{0\} \cup (0, \overline{\beta}_c), \\ W_\infty = 0 \text{ a.e. if } \beta > \overline{\beta}_c. \end{cases}$$

Some closing remarks:

(i) $\overline{\beta}_c \geq \beta \Rightarrow W_\infty > 0 \Rightarrow p = \lambda \Rightarrow \beta_c \geq \beta$
$\Rightarrow \beta_c \geq \overline{\beta}_c$. Does $\leq$ hold?

(ii) $W_\infty = 0$ near $\beta_0 \Rightarrow p(\beta) < \lambda(\beta)$ near β_0, in some window?

(iii) What is W_∞ when $\beta = \overline{\beta}_c$?

· **Conjecture** (true in $d = 1, 2$): $\beta_c = \overline{\beta}_c$,

and $W_\infty(\overline{\beta}_c) = 0$.